

CUXHAVEN, Germany

WMO: 101310

Lat: **53.8714N**Lon: **8.7058E**Elev: **12**StdP: **101.18**Time Zone: **1.00 (EUC)**Period: **82-03**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.4	-6.3	-11.2	1.4	-7.3	-9.4	1.7	-5.8	15.8	4.4	14.5	2.1	5.9	90	0.693

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.4	25.2	18.8	23.3	18.1	21.6	17.3	19.7	23.6	18.8	22.2	17.9	20.9	5.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.1	21.6	17.4	12.4	20.7	16.5	11.8	19.8	56.2	23.8	53.3	22.2	50.5	20.9	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.3	11.8	10.5	-8.2	29.9	4.4	2.8	-11.3	31.9	-13.9	33.5	-16.4	35.1	-19.6	37.1		
(5)			-8.8	22.3	4.1	1.0	-11.7	23.0	-14.1	23.5	-16.4	24.1	-19.4	24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	1.9	1.6	4.5	7.6	12.1	15.0	17.1	17.0	14.3	10.6	5.9	3.6
	DBStd	6.39	4.52	4.03	3.22	2.86	3.27	3.06	2.53	2.44	2.04	2.89	3.00	3.39
	HDD10.0	1104	251	234	172	83	14	1	0	0	0	27	124	199
	HDD18.3	3336	509	467	429	322	196	109	56	57	123	239	372	457
	CDD10.0	856	0	1	2	11	78	150	221	218	128	46	2	1
	CDD18.3	46	0	0	0	0	2	9	17	16	2	0	0	0
	CDH23.3	168	0	0	0	1	11	27	58	66	4	0	0	0
	CDH26.7	24	0	0	0	0	1	2	8	13	0	0	0	0
Wind	WSAvg	5.6	6.4	6.2	5.8	5.3	5.4	5.0	5.1	4.9	5.2	5.7	5.8	6.0
Precipitation	PrecAvg	853	66	52	50	40	57	84	92	91	85	82	79	75
	PrecMax	1140	140	116	93	97	116	182	182	216	200	201	164	189
	PrecMin	499	1	18	9	6	12	7	39	27	27	27	10	17
	PrecStd	141	34	26	25	20	25	43	43	46	46	44	38	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.8	13.3	15.3	18.5	24.7	26.3	27.5	28.2	23.5	19.7	13.5	11.9
		MCWB	9.3	10.8	11.0	12.5	16.9	19.3	20.2	20.4	18.2	16.0	11.4	10.0
	2%	DB	9.1	10.4	11.5	15.8	21.9	23.7	24.9	25.0	20.3	17.5	12.0	10.5
		MCWB	8.1	8.3	9.1	11.0	16.1	17.9	19.2	19.0	16.3	14.5	10.7	9.2
	5%	DB	8.1	8.2	9.9	13.7	19.3	21.5	22.9	22.6	18.6	15.6	11.0	9.2
		MCWB	7.2	6.7	7.9	10.3	14.7	17.0	18.3	18.1	15.7	13.5	9.9	8.1
	10%	DB	7.2	6.7	8.6	12.0	17.0	19.7	21.1	20.7	17.4	14.2	10.0	8.1
		MCWB	6.2	5.3	7.1	9.3	13.6	16.1	17.4	17.3	14.8	12.5	9.0	7.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	11.1	11.4	13.6	17.6	20.4	21.1	21.0	18.8	16.8	12.0	10.5
		MCDB	10.4	13.2	14.9	17.1	22.9	24.8	25.2	25.9	21.7	19.2	13.0	11.4
	2%	WB	8.1	8.6	9.4	11.7	16.3	18.6	19.7	19.5	17.4	15.0	10.8	9.4
		MCDB	8.8	10.1	10.9	14.5	21.2	21.8	23.2	23.9	19.7	16.8	11.8	10.4
	5%	WB	7.3	6.9	8.2	10.7	15.0	17.5	18.9	18.6	16.2	13.8	10.0	8.3
		MCDB	8.0	8.2	9.6	13.3	18.7	20.6	22.2	22.1	18.1	15.2	10.8	9.0
	10%	WB	6.3	5.5	7.3	9.6	13.8	16.5	17.9	17.7	15.3	12.9	9.1	7.3
		MCDB	7.1	6.5	8.5	11.6	16.9	19.6	20.8	20.4	16.9	14.0	9.9	8.0
Mean Daily Temperature Range	5% DB	MDBR	3.6	3.9	4.6	5.7	5.9	5.2	5.4	5.6	4.9	4.6	3.9	3.4
		MCDBR	3.9	5.7	6.8	9.0	9.9	8.8	8.9	9.7	7.5	7.0	5.1	4.6
		MCWBR	3.7	4.6	5.0	5.7	5.7	5.1	4.9	5.2	4.8	4.7	4.5	4.2
	5% WB	MCDBR	4.0	5.5	6.3	8.2	9.4	7.7	8.0	8.1	6.6	5.8	4.9	4.3
MCWBR		3.8	4.7	5.0	5.6	5.5	5.0	5.0	4.8	4.9	4.4	4.4	4.1	
Clear-Sky Solar Irradiance	taub	0.328	0.346	0.375	0.408	0.401	0.400	0.416	0.411	0.387	0.363	0.341	0.319	
	taud	2.309	2.326	2.270	2.236	2.306	2.334	2.302	2.326	2.378	2.398	2.348	2.304	
	Ebn at Noon	621	740	796	818	846	850	826	807	779	712	600	550	
	Edh at Noon	60	80	104	123	120	118	120	111	94	74	58	50	
All-Sky Solar Radiation	RadAvg	0.55	1.21	2.43	4.04	5.08	5.29	5.02	4.13	2.98	1.64	0.69	0.41	
	RadStd	0.06	0.17	0.24	0.47	0.54	0.43	0.64	0.31	0.34	0.18	0.07	0.03	